Valid, satisfiable, and unsatisfiable sets of formulas

Definition (Model of a set of formula)

An interpretation $\mathcal{I}$ that satisfies all the formulas of a set Γ, i.e., such that $\mathcal{I} \models A$ for all formulas $A \in \Gamma$, is called a model of Γ (written $\mathcal{I} \models \Gamma$).

Definition

A set of formulas Γ is

- **Valid** if for all $\mathcal{I}$, $\mathcal{I} \models \Gamma$, i.e., every interpretation is a model of Γ
- **Satisfiable** if there is an $\mathcal{I}$ s.t. $\mathcal{I} \models \Gamma$, i.e., Γ has at least one model
- **Unsatisfiable** if for no $\mathcal{I}$, $\mathcal{I} \models \Gamma$, i.e., Γ has no model

Proposition

For any finite set of formulas Γ, (i.e., $\Gamma = \{A_1, \ldots, A_n\}$ for some $n \geq 1$), Γ is valid (resp. satisfiable and unsatisfiable) if and only if $A_1 \wedge \cdots \wedge A_n$ is valid (resp, satisfiable and unsatisfiable).

Validity check by truth tables: Example

We use the truth tables method to determine whether $(p \to q) \lor (p \to \neg q)$ is valid.

p	q	$p \to q$	$\neg q$	$p \to \neg q$	$(p \to q) \lor (p \to \neg q)$
True	True	True	False	False	True
True	False	False	True	True	True
False	True	True	False	True	True
False	False	True	True	True	True

The formula is valid since it is satisfied by every interpretation.

Satisfiablity check by truth tables: Example

Use the truth tables method to determine whether $(\neg p \lor q) \land (q \to \neg r \land \neg p) \land (p \lor r)$ (denoted with ϕ) is satisfiable.

p	q	r	$\neg p \lor q$	$\neg r \land \neg p$	$q \to \neg r \land \neg p$	$(p \lor r)$	ϕ
True	True	True	True	False	False	True	False
True	True	False	True	False	False	True	False
True	False	True	False	False	True	True	False
True	False	False	False	False	True	True	False
False	True	True	True	False	False	True	False
False	True	False	True	True	True	False	False
False	False	True	True	False	True	True	True
False	False	False	True	True	True	False	False

There exists an interpretation satisfying ϕ, thus ϕ is satisfiable.

Truth tables: Exercises

Use the truth table method to verify whether the following formulas are valid, satisfiable or unsatisfiable:

- $(p \to q) \wedge \neg q \to \neg p$
- $(p \to q) \to (p \to \neg q)$
- $(p \vee q \to r) \vee p \vee q$
- $(p \vee q) \wedge (p \to r \wedge q) \wedge (q \to \neg r \wedge p)$
- $(p \to (q \to r)) \to ((p \to q) \to (p \to r))$
- $(p \vee q) \wedge (\neg q \wedge \neg p)$
- $(\neg p \to q) \vee ((p \wedge \neg r) \leftrightarrow q)$
- $(p \to q) \wedge (p \to \neg q)$
- $(p \to (q \vee r)) \vee (r \to \neg p)$

Logical consequence (or logical implication)

Definition (Logical consequence or logical implication)

A formula A is a logical consequence of (or, is logically implied by) a set of formulas Γ, denoted by $\Gamma \models A$ if every model of Γ is also a model of A (i.e., for all $\mathcal{I}$, $\mathcal{I} \models \Gamma$ implies $\mathcal{I} \models A$.

We write $\models A$ to mean $\emptyset \models A$, i.e., to mean that A is valid. We write $\Gamma \not\models A$ to mean that A is not logically implied by Γ. Thus, $\not\models A$ means A not valid.

Observation

$\Gamma \not\models A$ means that there exists at least one model of Γ where A is false.

If Γ_1 and Γ_2 are sets of formulas, we also define $\Gamma_1 \models \Gamma_2$ simply as $\Gamma_1 \models A$ for every $A \in \Gamma_2$. Thus, $\emptyset \models \Gamma$, $\not\models \Gamma$, and $\Gamma_1 \not\models \Gamma_2$ are defined accordingly.

Overloading of $\models$

Note that the symbol $\models$ is overloaded:

- when $\models$ has an intepretation $\mathcal{I}$ on the left, then it means satisfaction, i.e., $\mathcal{I} \models \alpha$ means that $\mathcal{I}$ satisfies α;
- when $\models$ has a formula Γ on the left, then it means logical implication, i.e., $\Gamma \models \alpha$ means that Γ logically implies α.

Logical equivalence

Definition (Logical equivalence)

Two formulas F and G are logically equivalent (denoted with $F \models\!\dashv G$) if both $F \models G$ and $G \models F$, i.e., for every interpretation $\mathcal{I}$, $\mathcal{I}(F) = \mathcal{I}(G)$ (in other words, if the set of models of F and G coincide).

Logical equivalence is naturally extended to sets of formulas: $\Gamma_1 \models\!\dashv \Gamma_2$ if both $\Gamma_1 \models \Gamma_2$ and $\Gamma_2 \models \Gamma_2$.

Sometimes, logical equivalence is denoted by the same symbol used for "material equivalence", i.e., $\equiv$. However, it is important not to confuse the two notions:

- material implication is in the language of propositonal logic
- logical implication is in the meta-language, i.e., it is outside the language of propositonal logic.

Exercises on logical implication

Check that the following logical implications hold

- $p \models p \vee q$
- $q \vee p \models p \vee q$
- $\{p \vee q, p \rightarrow r, q \rightarrow r\} \models r$
- $\{p \rightarrow q, p\} \models q$

Hint: base your check on the definition of "satisfaction", i.e., check that every model of the formulas on the left is also a model of the formula on the right.

Solution: checking logical consequence in a direct manner

Verify that $p \models p \vee q$ Suppose that $\mathcal{I} \models p$, then by definition $\mathcal{I} \models p \vee q$.

Verify that $q \vee p \models p \vee q$ Suppose that $\mathcal{I} \models q \vee p$, then $\mathcal{I} \models q$ or $\mathcal{I} \models p$. In both cases we have that $\mathcal{I} \models p \vee q$.

Verify that $\{p \vee q, p \rightarrow r, q \rightarrow r\} \models r$ Suppose that $\mathcal{I} \models p \vee q$ and $\mathcal{I} \models p \rightarrow r$ and $\mathcal{I} \models q \rightarrow r$. Then $\mathcal{I} \models p$ or $\mathcal{I} \models q$. In the first case, since $\mathcal{I} \models p \rightarrow r$, then $\mathcal{I} \models r$. In the second case, since $\mathcal{I} \models q \rightarrow r$, then $\mathcal{I} \models r$.

Verify that $\{p \rightarrow q, p\} \models q$ Suppose that $\mathcal{I} \models p \rightarrow q$ and $\mathcal{I} \models p$. By the semantics of $\rightarrow$, $\mathcal{I} \models p \rightarrow q$ means that $\mathcal{I} \models \neg p \vee q$. Therefore, $\mathcal{I} \models q$.

Logical consequence and equivalence using truth tables

Use the truth tables method to determine whether $\neg p \models p \wedge \neg q \to p \wedge q$.

p	q	$\neg p$	$p \wedge \neg q$	$p \wedge q$	$p \wedge \neg q \to p \wedge q$
True	True	False	False	True	True
True	False	False	True	False	False
False	True	True	False	False	True
False	False	True	False	False	True

Use the truth tables method to determine whether $p \to (q \wedge \neg q) \models\!\!\!\dashv \neg p$.

p	q	$q \wedge \neg q$	$p \to (q \wedge \neg q)$	$\neg p$
True	True	False	False	False
True	False	False	False	False
False	True	False	True	True
False	False	False	True	True

Truth tables: exercises

Use the truth table method to determine whether the following logical consequences and logical equivalences hold:

- $(p \to q) \models \neg p \to \neg q$
- $(p \to q) \land \neg q \models \neg p$
- $p \to q \land r \models (p \to q) \to r$
- $p \lor (\neg q \land r) \models q \lor \neg r \to p$
- $\neg(p \land q) \models\!\!\!\Dashv \neg p \lor \neg q$
- $(p \lor q) \land (\neg p \to \neg q) \models\!\!\!\Dashv q$
- $(p \land q) \lor r \models\!\!\!\Dashv (p \to \neg q) \to r$
- $(p \lor q) \land (\neg p \to \neg q) \models\!\!\!\Dashv p$
- $((p \to q) \to q) \to q \models\!\!\!\Dashv p \to q$

Properties of propositional logical consequence

Proposition

If Γ and Σ are two sets of propositional formulas, and A and B two formulas, then the following properties hold:

- **Reflexivity** *If $A \in \Gamma$, then $\Gamma \models A$*
- **Ex falso sequitur quodlibet** *If Γ is unsatisfiable, then $\Gamma \models A$ for all A*
- **Monotonicity** *If $\Gamma \models A$ then $\Gamma \cup \Sigma \models A$*
- **Cut** *If $\Gamma \models A$ and $\Sigma \cup \{A\} \models B$ then $\Gamma \cup \Sigma \models B$*
- **Compactness** *If $\Gamma \models A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \models A$*
- **Deduction theorem** *If $\Gamma \cup \{A\} \models B$ then $\Gamma \models A \rightarrow B$*
- **Deduction principle** *$\Gamma \cup \{A\} \models B$ if and only if $\Gamma \models A \rightarrow B$*
- **Refutation principle** *$\Gamma \models A$ if and only if $\Gamma \cup \{\neg A\}$ is unsatisfiable*

Properties of propositional logical consequence

Reflexivity If $A \in \Gamma$, then $\Gamma \models A$.
PROOF: If $A \in \Gamma$, and $\mathcal{I} \models \Gamma$, then obviously $\mathcal{I} \models A$.

Ex falso sequitur quodlibet If Γ is unsatisfiable, then $\Gamma \models A$ for all A
PROOF: Recall that $\Gamma \not\models A$ means that there is a model of Γ where A is false. But if Γ is unsatisfiable there is no model of Γ, and therefore it cannot be that $\Gamma \not\models A$. Thus we conclude that $\Gamma \models A$.

Monotonicity If $\Gamma \models A$ then $\Gamma \cup \Sigma \models A$
PROOF: Every model of $\Gamma \cup \Sigma$ is clearly a model Γ. Therefore, If $\Gamma \models A$, then every model of $\Gamma \cup \Sigma$ is also a model of A, i.e., $\Gamma \cup \Sigma \models A$.

Cut If $\Gamma \models A$ and $\Sigma \cup \{A\} \models B$ then $\Sigma \cup \Gamma \models B$.
PROOF: For all $\mathcal{I}$, if $\mathcal{I} \models \Gamma \cup \Sigma$, then $\mathcal{I} \models \Gamma$ and $\mathcal{I} \models \Sigma$. The hypothesis $\Gamma \models A$ implies that $\mathcal{I} \models A$. Since $\mathcal{I} \models \Sigma$, then $\mathcal{I} \models \Sigma \cup \{A\}$. The hypothesis $\Sigma \cup \{A\} \models B$, implies that $\mathcal{I} \models B$. We can therefore conclude that $\Sigma \cup \Gamma \models B$.

Properties of propositional logical consequence

Compactness If $\Gamma \models A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \models A$.

PROOF: Let $\mathcal{P}_A$ be the propositional variables occurring in A. Let $\mathcal{I}_1, \dots, \mathcal{I}_n$ (with $n \leq 2^{|\mathcal{P}_A|}$), be all the interpretations of the language $\mathcal{P}_A$ that do not satisfy A. Since $\Gamma \models A$, then there should be $\mathcal{I}'_1, \dots, \mathcal{I}'_n$ interpretations of the language of Γ, which are extensions of $\mathcal{I}_1, \dots, \mathcal{I}_n$, and such that $\mathcal{I}'_k \not\models \gamma_k$ for some $\gamma_k \in \Gamma$. Let $\Gamma_0 = \{\gamma_1, \dots, \gamma_k\}$. Then $\Gamma_0 \models A$. Indeed if $\mathcal{I} \models \Gamma_0$ then $\mathcal{I}$ is an extension of an interpretation J of $\mathcal{P}_A$ that satisfies A, and therefore $\mathcal{I} \models A$.

Deduction theorem If $\Gamma \cup \{A\} \models B$ then $\Gamma \models A \rightarrow B$

PROOF: Assume $\Gamma \cup \{A\} \models B$. Consider an interpretation $\mathcal{I}$, and suppose that $\mathcal{I} \models \Gamma$. If $\mathcal{I} \not\models A$, then $\mathcal{I} \models A \rightarrow B$. If instead $\mathcal{I} \models A$, then $\mathcal{I} \models \Gamma \cup \{A\}$. But since $\Gamma \cup \{A\} \models B$, we conclude that $\mathcal{I} \models B$. We can therefore conclude that $\Gamma \models A \rightarrow B$.

Properties of propositional logical consequence

Deduction principle $\Gamma \cup \{A\} \models B$ if and only if $\Gamma \models A \rightarrow B$

PROOF: One direction coincides with the deduction theorem. The other direction amount to prove that if $\Gamma \models A \rightarrow B$, then $\Gamma \cup \{A\} \models B$. But $\Gamma \models A \rightarrow B$ means that every model of Γ is a model of $\neg A \vee B$. This means that every model of Γ where A is true must have B true, and thus every model of $\Gamma \cup A$ is a model of B, which means $\Gamma \cup \{A\} \models B$.

Refutation principle $\Gamma \models A$ if and only if $\Gamma \cup \{\neg A\}$ is unsatisfiable

PROOF:

$(\Longrightarrow)$ Suppose by contradiction that $\Gamma \cup \{\neg A\}$ is satisfiable. This implies that there is an interpretation $\mathcal{I}$ such that $\mathcal{I} \models \Gamma$ and $\mathcal{I} \models \neg A$, i.e., $\mathcal{I} \not\models A$. This contradicts that fact that for all interpretations that satisfies Γ, they satisfy A

$(\Longleftarrow)$ Let $\mathcal{I} \models \Gamma$, then by the fact that $\Gamma \cup \{\neg A\}$ is unsatisfiable, we have that $\mathcal{I} \not\models \neg A$, and therefore $\mathcal{I} \models A$. We can conclude that $\Gamma \models A$.